

Aditya Bhosale

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Education

University of Illinois Urbana-Champaign

Aug 2021 – Feb 2027 (Expected)

Ph.D. in Computer Science, Advisor: Laxmikant Kale

Research Interests: Parallel Programming, Cloud Computing, Distributed Runtime Systems

Indian Institute of Technology, Bombay

Jul 2014 – Aug 2019

Dual Degree (B.Tech. & M.Tech.) in Aerospace Engineering, Advisor: Prabhu Ramachandran

Master's Thesis: A generalized framework for heterogeneous computing

Bachelor's Thesis: Optimal nearest neighbor searching algorithms for particle-based simulations

Professional Experience

Uber

Sunnyvale

PhD Software Engineer Intern

May 2026 – Aug 2026

- Built a hardware performance monitoring pipeline collecting performance counters across millions of containers on 100K+ hosts.
- Developed service-latency optimizations from hardware performance data, including topology-aware CPU affinity.

NVIDIA

Santa Clara

HPC Architect Intern

May 2024 – Aug 2024

- Developed performance models for rapid runtime prediction of ML workloads from GPU compute and network characteristics.

Lawrence Livermore National Laboratory

Remote

Student Intern

May 2023 – Dec 2023

- Investigated machine-learning representations of LLVM IR for graph neural network models.
- Built a Record-Replay dataset for learning optimal OpenMP GPU-offload launch configurations.

Goldman Sachs

India

Associate

May 2019 – Aug 2021

- Managed cloud infrastructure for a market-risk pricing framework, contributing to a 10% reduction in VaR compute cost through improved load balancing and runtime prediction.
- Received a fast-track promotion from Analyst to Associate in 1.5 years.

Selected Research

Distributed GPU Runtime Systems

- Designed a hierarchical execution model to amortize host-side scheduling and launch overheads for fine-grained, overdecomposed multi-GPU workloads.
- Designed a GPU communication protocol supporting asynchronous communication among migratable actors.
- Developed asynchronous, split-phase GPU load balancing with SM-utilization-aware load measurement and deadlock-free migration scheduling under GPU memory constraints.
- Developed an online model for learning kernel-level GPU efficiency to enable load balancing across heterogeneous GPUs.
- Implemented restart-free, checkpoint-free resource elasticity in Charm++ through dynamic reconfiguration of low-level communication endpoints.
- Enabled HPC and ML applications to execute on cloud spot GPUs with less than 10 ms of runtime overhead for handling spot interruptions.

Adaptive HPC DSLs

- Designed and developed CharmTyles, a client-server architecture for cloud-native HPC DSLs with Python frontends and a distributed Charm++ backend, including reusable support for asynchronous lazy evaluation, operator fusion, dependency tracking, and garbage collection.
- Developed a portable dense linear algebra DSL for distributed GPUs with MLIR-based JIT compilation.
- Developed an adaptive distributed GPU stencil DSL with kernel fusion and JIT-generated CUDA kernels, outperforming cuPyNumeric and Devito.
- Developed a Pandas-like adaptive, in-memory distributed data analytics abstraction using Charm++ and Apache Arrow.

Publications

- **Efficient and Portable Support for Overdecomposition on Distributed GPU Platforms.** 2026
Aditya Bhosale, et al.
PAW-ATM (SC Workshops '26). [paper]
- **No-Restart Elasticity in an Adaptive Runtime System for Cloud-Native HPC.** 2026
Aditya Bhosale, Laxmikant Kale.
CANOPIE-HPC (SC Workshops '26). [paper]
- **Towards an Adaptive Runtime System for Cloud-Native HPC.** 2026
Aditya Bhosale, Advait Tahilyani, Laxmikant Kale, and Sara Kokkila-Schumacher.
AsHES (IPDPS Workshops '26). [paper]
- **An Adaptive Distributed Stencil Abstraction for GPUs.** 2025
Aditya Bhosale, Laxmikant Kale.
Supercomputing India 2025 Conference, Bangalore, India. [paper]
- **An Elastic Job Scheduler for HPC Applications on the Cloud.** 2025
Aditya Bhosale, Kavitha Chandrasekar, Laxmikant Kale, Sara Kokkila-Schumacher.
CANOPIE-HPC (SC Workshops '25), St. Louis, MO. [paper]
- **Efficient and Cost-Effective HPC on the Cloud.** 2025
Aditya Bhosale, Laxmikant Kale, Sara Kokkila-Schumacher.
FlexScience (HPDC Workshops '25), Notre Dame, IN. [paper]
- **An Abstraction for Distributed Stencil Computations Using Charm++.** 2024
Aditya Bhosale, Zane Fink, Laxmikant Kale.
Workshop on Asynchronous Many-Task Systems and Applications, Knoxville, TN. [paper]
- **Runtime Techniques for Automatic Process Virtualization.** 2022
Evan Ramos, Sam White, Aditya Bhosale, Laxmikant Kale.
P2S2 (ICPP Workshops '22), Bordeaux, France. [paper]
- **Compyle: a Python package for parallel computing.** 2020
Aditya Bhosale, Prabhu Ramachandran.
19th Python in Science Conference, Austin, TX. [paper]
- **PySPH: A Python-Based Framework for Smoothed Particle Hydrodynamics.** 2019
Prabhu Ramachandran, Aditya Bhosale, et al.
ACM Transactions on Mathematical Software. [paper]

Selected Talks

- 2024:** Aditya Bhosale. *Charm++ on Kubernetes Cloud*. Talk, KubeCon 2024, Salt Lake City.
- 2023:** Aditya Bhosale, Nikunj Gupta, Zane Fink. *CharmTypes: Large-scale Interactive Charm++ with Python*. Talk, 15th JLESC Workshop, INRIA, Bordeaux.
- 2023:** Aditya Bhosale, Kavitha Chandrasekar, et al. *Support for Charm++ on Cloud using Kubernetes*. Talk, 15th JLESC Workshop, INRIA, Bordeaux.

Awards

- 2020:** Best Poster Award, SciPy 2020 – *Compyle: Python Once, Parallel Computing Anywhere*
- 2016:** Undergraduate Research Award, IIT Bombay

Technical Skills

Languages: C++, Python, Go

GPU/Parallel: CUDA, HIP, OpenCL, MPI, Charm++, OpenMP, Kokkos

Compilers: MLIR, LLVM

Cloud/Infrastructure: Kubernetes